

What is kubectl api-resources ?

- `kubectl api-resources` This command lists all available resource types in your cluster.
- API resources are the object types available in Kubernetes
- It lists all available resource types supported by the Kubernetes API server.
- 👉 It shows:
 - Resource name
 - Short name
 - API group
 - Namespaced or not
 - Kind

What is the importance of API resources in Kubernetes ?

- They are the foundation of Kubernetes.
- 👉 Everything in Kubernetes is an API object:
 - You create resources
 - Kubernetes maintains desired state
 - Controllers manage them

Can we extend Kubernetes API ?

- Yes, using: CRDs

Difference between Resource and Kind ?

 Term	 Meaning
 Resource	API endpoint (e.g., pods , deployments)
 Kind	Object type in YAML (e.g., Pod , Deployment)

What are namespaced vs non-namespaced resources ?

- 👉 Check: `kubectl api-resources --namespaced=true`

 Type	 Description	 How It Works	 Examples
 Namespaced	Exist within a namespace	👉 Created and accessed inside a specific namespace (default , dev , prod)	Pods, Deployments, Services, ConfigMaps
 Cluster-wide	Exist at cluster level	👉 Not tied to any namespace → available across cluster	Nodes, PersistentVolumes (PV), StorageClasses

What is an API Group ?

- API groups organize resources logically.
- 👉 Examples:
 - v1 → Core group
 - apps/v1 → Deployments
 - batch/v1 → Jobs

What are Custom Resource Definitions (CRDs) ?

- CRDs allow you to create your own API resources in Kubernetes.
- 👉 Example:
 - Prometheus creates ServiceMonitor
 - ArgoCD creates Application

Scenario : You need to list only cluster-level resources

How do you find resources that are NOT namespaced ?

- `kubectl api-resources --namespaced=false`
- 👉 Examples: PersistentVolumes , StorageClasses

Scenario : Too many resources in cluster

Your cluster shows 200+ resources. Why ?

-  Many CRDs installed
- generated by "Markdown to PDF Fast Converter"  <https://dub.sh/79wTuSy>

- Tools like : Prometheus , Istio , ArgoCD
- 👉 Each tool adds custom resources

⚡ Kubernetes API Resources – Rapid Fire Q&A

 #	? Question	💡 Answer
♦ 1	What is a Kubernetes API Resource?	👉 Any object you interact with via the Kubernetes API (Pods , Deployments , Services , etc.).
♦ 2	Where are API resources defined?	👉 In the Kubernetes API server .
♦ 3	Command to list all API resources?	👉 <code>kubectl api-resources</code>
♦ 4	Command to get API versions?	👉 <code>kubectl api-versions</code>
🧠 5	What is an API Group?	👉 Logical grouping of resources (e.g., apps , batch).
🧠 6	What is API Version?	👉 Version of resource schema (e.g., v1 , apps/v1).
🧠 7	What is Kind?	👉 Type of resource (e.g., Pod , Deployment).
🧠 8	What is Resource vs Kind?	👉 Resource = API endpoint (pods), Kind = object type (Pod).
📦 9	Core group examples?	👉 Pods , Services , ConfigMaps , Secrets (v1)
📦 10	apps/v1 group?	👉 Deployments , StatefulSets , DaemonSets
📦 11	batch/v1 group?	👉 Jobs , CronJobs
📦 12	networking.k8s.io/v1?	👉 Ingress , NetworkPolicy
 13	Main sections of a resource YAML?	👉 apiVersion, kind, metadata, spec, status
 14	What is metadata?	👉 Info like name , labels , namespace .
 15	What is spec?	👉 Desired state.
 16	What is status?	👉 Current state (managed by Kubernetes).

 #	? Question	💡 Answer
 17	Are all resources namespaced?	👉 ❌ No
 18	Examples of cluster-wide resources?	👉 Nodes , PersistentVolumes , Namespaces
 19	How to create a resource?	👉 <code>kubectl apply -f file.yaml</code>
 20	How to delete a resource?	👉 <code>kubectl delete -f file.yaml</code>
 21	How to edit a resource?	👉 <code>kubectl edit</code>
 22	How to inspect a resource definition?	👉 <code>kubectl explain <resource></code>
 23	How to get detailed info?	👉 <code>kubectl get <resource> -o yaml</code>
 24	What is CRD?	👉 Custom Resource Definition — extends Kubernetes API .
 25	Why use CRDs?	👉 To define custom objects (e.g., monitoring , databases).
 26	Example CRD usage?	👉 Istio resources like VirtualService
 27	What is a controller?	👉 Watches resources and ensures desired state .
 28	Example?	👉 Deployment controller ensures pods match replica count .
 29	How does kubectl interact with API?	👉 Via API server using REST .
 30	What is kubeconfig ?	👉 Config file for cluster access .
 31	What are API stability levels?	👉 Alpha , Beta , Stable (GA)
 32	Can API versions be deprecated?	👉 ✅ Yes
 33	What is resource quota?	👉 Limits resource usage in namespace .
 34	What is limit range?	👉 Default/min/max resource constraints.

 #	? Question	💡 Answer
 35	What is field selector?	👉 Filter resources by fields .
 36	Why is apiVersion important?	👉 Ensures compatibility with cluster version .
 37	What happens if wrong apiVersion used?	👉 Resource creation fails .
 38	How to find correct API version?	👉 <code>kubectl api-resources</code>
 39	Why CRDs are powerful?	👉 Extend Kubernetes into platform (operators).